# LeetCode & HackerRank Solutions

LeetCode & HackerRank Assigned Problems

Name: Muhammad Farhan
Roll no: 25K-0538
Section: BCS-1K
Course: Programming Fundamentals Lab
Course Instructor: Sir Sheeraz Iqbal

70. Climbing stairs

Accepted 45 / 45 testcases passed

Farhan submitted at Nov 23, 2025 11:56

Editorial Solution

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Runtime

0 ms Beats 100.00%

Analyze Complexity

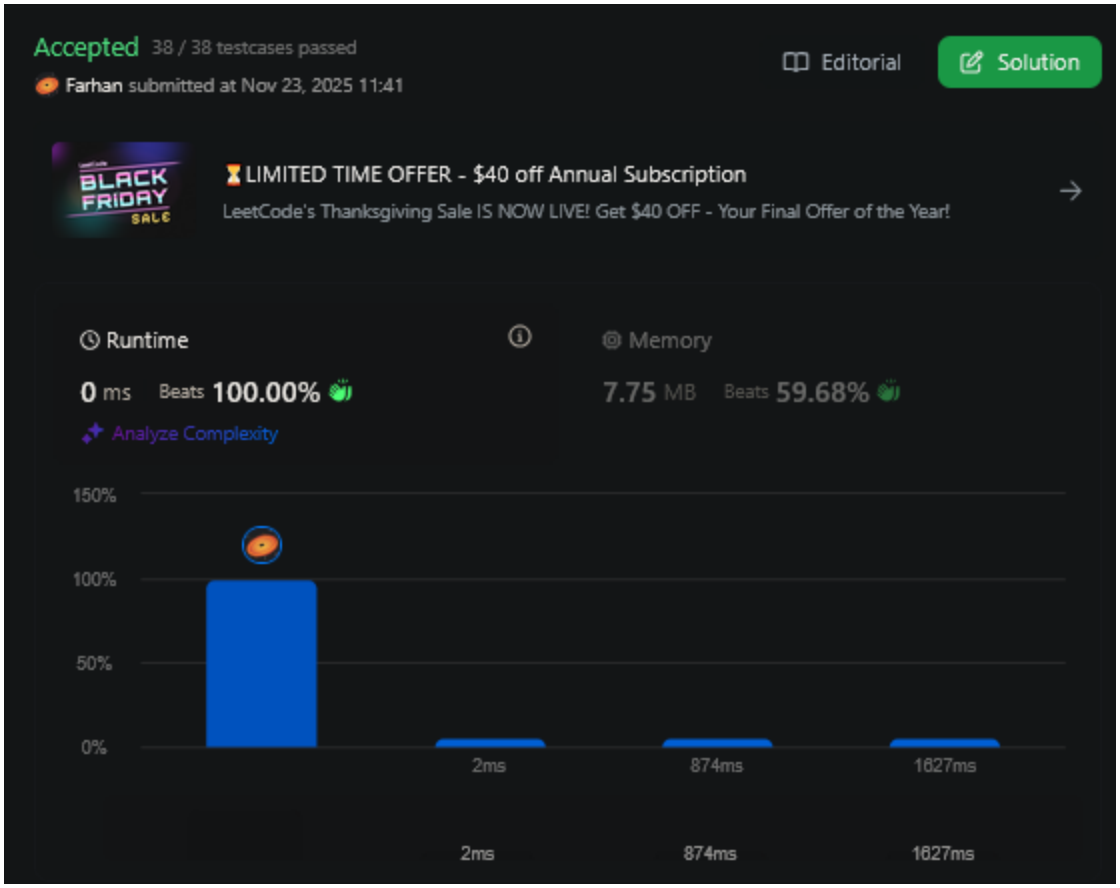
Memory

7.74 MB Beats 63.21%

Code C

```c
int book[50] = {0};
int climbStairs(int n) {
    if (n<=2)
    {
        return n;
    }
    if (book[n]!=0)
    {
        return book[n];
    }
    book[n] = climbStairs(n-1)+climbStairs(n-2);
    return book[n];
}
```

1137. N-th Tribonacci Number




Code C

```c
int book[40] = {0};
int tribonacci(int n) {
    if (n<=0)
    {
        return 0;
    }
    if (n==1 || n==2)
    {
        return 1;
    }
    if (book[n]!=0)
    {
        return book[n];
    }
    book[n] = tribonacci(n-3)+tribonacci(n-2)+tribonacci(n-1);
    return book[n];
}
```

View less

HackerRank: **Struct**

Change Theme  Language C++11

```cpp
#include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>
#include <algorithm>
using namespace std;
/*
Name: Muhammad Farhan
Roll no: 25K-0538
Section: BCS-1K
*/
struct Student
{
    int age;
    string first_name;
    string last_name;
    int standard;
};
int main() {
    Student st;

    cin >> st.age >> st.first_name >> st.last_name >> st.standard;
    cout << st.age << " " << st.first_name << " " << st.last_name << " " << st.standard;

    return 0;
}

```

You made this submission 6 minutes ago.

**Score:** 10.00  **Status:** Accepted

People who solved **Structs** attempted this next:

## Class

Learn how to create and use classes.

Solve Challenge

### Submitted Code

**Language:** C++  Open in editor

```cpp
#include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>
#include <algorithm>
using namespace std;
/*
Name: Muhammad Farhan
Roll no: 25K-0538
Section: BCS-1K
*/
struct Student
{
    int age;
    string first_name;
```

## Submitted Code

**Language:** C++ Open in editor

```cpp
struct Student
{
    int age;
    string first_name;
    string last_name;
    int standard;
};
int main() {
    Student st;

    cin >> st.age >> st.first_name >> st.last_name >> st.standard;
    cout << st.age << " " << st.first_name << " " << st.last_name << " " << st.standard;

    return 0;
}
```

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case 5

Compiler Message

```
Success
```

Input (stdin) Download

```
15
john
carmack
10
```

Expected Output Download

```
15 john carmack 10
```

HackerRank: **Boxes through a Tunnel**

Change Theme  Language: C

```c
#include <stdio.h> ···
/*
Name: Muhammad Farhan
Roll no: 25K-0538
Section: BCS-1K
*/
struct box
{
    int width;
    int height;
    int length;
};

typedef struct box box;

int get_volume(box b) {
    return b.width*b.height*b.length;
}

int is_lower_than_max_height(box b) {
    if (b.height < 41)
        return 1;
    else
        return 0;
}

int main() ···
```

Problem | Submissions | Leaderboard | Discussions | Editorial

You made this submission 9 minutes ago.

**Score:** 25.00  **Status:** Accepted

People who solved **Boxes through a Tunnel** attempted this next:

**Small Triangles, Large Triangles**

Sort triangles by area

Solve Challenge

**Submitted Code**

**Language:** C  Open in editor

```c

/*
Name: Muhammad Farhan
Roll no: 25K-0538
Section: BCS-1K
*/
struct box
{
    int width;
    int height;
    int length;
};

typedef struct box box;
```

## Submitted Code

**Language:** C

Open in editor

```c
typedef struct box box;

int get_volume(box b) {
    return b.width*b.height*b.length;
}

int is_lower_than_max_height(box b) {
    if (b.height < 41)
        return 1;
    else
        return 0;
}
```

Test case 0

Test case 1

Test case 2

Test case 3

Test case 4

Test case

Compiler Message

Success

Input (stdin) Download

```
4
5 5 5
1 2 40
10 5 41
7 2 42
```

Expected Output Download

```
125
80
```